

Chapter 1: Network Routing Protocols

This chapter introduces network routing protocols and explains why the technique matters in production systems. Section 1 covers the trade-offs engineers weigh when they adopt network routing protocols at scale, together with the failure modes that appear on page 1 of this handbook.

Chapter 2: Relational Database Indexing

This chapter introduces relational database indexing and explains why the technique matters in production systems. Section 2 covers the trade-offs engineers weigh when they adopt relational database indexing at scale, together with the failure modes that appear on page 2 of this handbook.

Chapter 3: Continuous Integration Pipelines

This chapter introduces continuous integration pipelines and explains why the technique matters in production systems. Section 3 covers the trade-offs engineers weigh when they adopt continuous integration pipelines at scale, together with the failure modes that appear on page 3 of this handbook.

Chapter 4: Container Orchestration

This chapter introduces container orchestration and explains why the technique matters in production systems. Section 4 covers the trade-offs engineers weigh when they adopt container orchestration at scale, together with the failure modes that appear on page 4 of this handbook.

Chapter 5: Public Key Cryptography

This chapter introduces public key cryptography and explains why the technique matters in production systems. Section 5 covers the trade-offs engineers weigh when they adopt public key cryptography at scale, together with the failure modes that appear on page 5 of this handbook.

Chapter 6: Distributed Consensus Algorithms

This chapter introduces distributed consensus algorithms and explains why the technique matters in production systems. Section 6 covers the trade-offs engineers weigh when they adopt distributed consensus algorithms at scale, together with the failure modes that appear on page 6 of this handbook.

Chapter 7: Compiler Optimisation Passes

This chapter introduces compiler optimisation passes and explains why the technique matters in production systems. Section 7 covers the trade-offs engineers weigh when they adopt compiler optimisation passes at scale, together with the failure modes that appear on page 7 of this handbook.

Chapter 8: Garbage Collection Strategies

This chapter introduces garbage collection strategies and explains why the technique matters in production systems. Section 8 covers the trade-offs engineers weigh when they adopt garbage collection strategies at scale, together with the failure modes that appear on page 8 of this handbook.

Chapter 9: Operating System Schedulers

This chapter introduces operating system schedulers and explains why the technique matters in production systems. Section 9 covers the trade-offs engineers weigh when they adopt operating system schedulers at scale, together with the failure modes that appear on page 9 of this handbook.

Chapter 10: Cache Coherence Protocols

This chapter introduces cache coherence protocols and explains why the technique matters in production systems. Section 10 covers the trade-offs engineers weigh when they adopt cache coherence protocols at scale, together with the failure modes that appear on page 10 of this handbook.

Chapter 11: Message Queue Delivery Guarantees

This chapter introduces message queue delivery guarantees and explains why the technique matters in production systems. Section 11 covers the trade-offs engineers weigh when they adopt message queue delivery guarantees at scale, together with the failure modes that appear on page 11 of this handbook.

Chapter 12: Load Balancing Algorithms

This chapter introduces load balancing algorithms and explains why the technique matters in production systems. Section 12 covers the trade-offs engineers weigh when they adopt load balancing algorithms at scale, together with the failure modes that appear on page 12 of this handbook.

Chapter 13: Observability And Structured Logging

This chapter introduces observability and structured logging and explains why the technique matters in production systems. Section 13 covers the trade-offs engineers weigh when they adopt observability and structured logging at scale, together with the failure modes that appear on page 13 of this handbook.

Chapter 14: Infrastructure As Code

This chapter introduces infrastructure as code and explains why the technique matters in production systems. Section 14 covers the trade-offs engineers weigh when they adopt infrastructure as code at scale, together with the failure modes that appear on page 14 of this handbook.

Chapter 15: Blue Green Deployment

This chapter introduces blue green deployment and explains why the technique matters in production systems. Section 15 covers the trade-offs engineers weigh when they adopt blue green deployment at scale, together with the failure modes that appear on page 15 of this handbook.

Chapter 16: Chaos Engineering Experiments

This chapter introduces chaos engineering experiments and explains why the technique matters in production systems. Section 16 covers the trade-offs engineers weigh when they adopt chaos engineering experiments at scale, together with the failure modes that appear on page 16 of this handbook.